

Talk Like a Structured Graph: Augmenting Text Encodings with Structural Statistics

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Abstract

Can natural-language descriptions of graph statistics improve a language model’s answers to graph queries? We prepend node degree, local clustering, or random-walk return statistics to an incident-text graph encoding and evaluate two Qwen3 models, each with plain and thinking prompts, on six tasks across several edge-density levels. Paired comparisons reveal gains in some conditions and losses in others; a statistic that helps one model or task can hinder another. Primer-only shortcut checks and explicit truncation reporting temper the interpretation of score gains. Overall, verbalized structural features do not provide a consistent improvement across the tested settings.

1 Introduction

Research question. When a black-box language model reads a textual graph, do explicit structural statistics improve execution of graph queries?

Graph neural networks and graph Transformers often receive explicitly constructed structural or positional features. These features can improve performance when the model consumes them numerically; GraphGPS, for example, incorporates random-walk structural encodings (RWSE) (Dwivedi et al., 2022; Rampásek et al., 2022). Their benefit in a learned graph architecture motivates testing whether a frozen language model can use related facts when they are written out as text.

As LLMs are increasingly asked to solve graph problems, GraphQA shows that the choice of text encoding can substantially improve results, but success still depends on the task and graph structure (Fatemi et al., 2024). NLGraph likewise finds uneven performance across graph problems (Wang et al., 2023). This makes explicit structural information a plausible complement to a textual edge description, while leaving open whether it improves the model’s use of the graph.

We test this question by adding degree, local clustering, and RWSE sentences before a fixed incident encoding and comparing the same graphs, questions, and models with and without those sentences. Related structural prompting exists (Huang et al., 2024); the specific gap here is a paired, task-by-task evaluation of these three statistics on GraphQA queries. A degree primer can already state a degree answer, or reveal the edge count through the degree sum. We therefore check what can be answered from the primer alone and separate answer disclosure from improvements on tasks requiring the edges. The results vary by task, feature, and model; no primer consistently improves performance.

2 Related Work

Structural encodings help trained graph models by giving them information their architectures may otherwise miss. Learnable structural and positional features improve some GNN benchmarks, while GraphGPS integrates numerical encodings such as RWSE with local message passing and global attention (Dwivedi et al., 2022; Rampásek et al., 2022). Our question differs: a frozen LLM sees rounded values as sentences before a graph description, without the learned pathway by which a graph model consumes a feature. Benefits in one setting therefore cannot be assumed to transfer to the other.

GraphQA and NLGraph establish that textual graph queries are sensitive to representation, task, and graph structure (Fatemi et al., 2024; Wang et al., 2023). We retain GraphQA’s incident encoding to isolate the addition of structural facts. Prior work has already put structural descriptions into LLM prompts for text-rich node classification, but finds that improved predictions need not mean the model interpreted topology as intended (Huang et al., 2024). Graph-description order can itself change answers (Ge et al., 2025); partial-input checks likewise expose shortcuts in model

evaluations (Feng et al., 2019). Together, these findings motivate paired query comparisons plus our filler and graph-blind controls. They leave open whether degree, clustering, and RWSE sentences help the same LLM answer the same GraphQA query when the graph encoding is held fixed.

3 Methodology

3.1 Models and graph corpus

Our preliminary evaluation used the published GraphQA graphs with 5–19 nodes and tested Gemma 4 E4B/12B and Qwen3-8B/14B, each with and without thinking. Most unprimed pilot arms were near ceiling on most evaluated tasks, leaving little room to measure a primer effect with the available graph samples. We therefore evaluated Qwen3-1.7B and Qwen3-4B, each in plain and native thinking mode, in the final paired study (Qwen Team, 2025). We increased the graph size to make baseline queries more demanding, while capping it at exactly 40 nodes because node-wise RWSE text and thinking responses consume substantial context and output tokens.

Our graphs use the undirected Erdős–Rényi generator family also available in GraphQA, with newly sampled instances rather than the original graphs (Fatemi et al., 2024; Gilbert, 1959). Every task is evaluated at edge probabilities $p \in \{.10, .20, .35, .50\}$; only node degree and edge existence have an additional extension at $p \in \{.65, .75, .85\}$. Each task–density level has 100 graphs, and the same instance is used for all primer conditions and model arms. The main sweep permits 8,192 output tokens; in the dense extension plain arms receive 2,048 and thinking arms 8,192. Different limits complicate between-density comparisons, so primer comparisons are made within a fixed arm, task, and graph set.

3.2 Tasks, primers, and paired prompts

We use the six GraphQA tasks (Fatemi et al., 2024): `node_degree`, `connected_nodes` (the queried node’s neighbors), `edge_count`, `node_count`, `cycle_check`, and `edge_existence`. Because every graph has 40 nodes, the node-count answer is fixed; every graph in the sampled corpus also contains a cycle. The resulting constant gold labels make these two tasks useful for observing prompt sensitivity but unable to establish discrimination between different answers.

Each prompt keeps the same incident graph encoding, node names, and question. We prepend either no primer, degree statements, clustering statements, RWSE statements, or all three types together. Two diagnostics sit outside this five-condition comparison: a sentence stating the graph’s component count and a *filler* preamble that names every node but gives no structural facts. The filler checks whether a longer preamble and repeated node names change answers even without useful graph information. Its mean length is 1,829 characters, near clustering’s 1,629, but wording and exact length are not matched, so it cannot isolate length alone. The primer is placed before the graph encoding, followed by the question.

For adjacency matrix A and transition matrix $P = D^{-1}A$ (zero rows at isolated vertices), the rendered per-node features are

$$d_v = \sum_u A_{vu}, \quad c_v = \frac{2t_v}{d_v(d_v - 1)}, \quad (1)$$

$$r_v^{(k)} = (P^k)_{vv}, \quad k \in \{2, 3\}. \quad (2)$$

Here t_v is the number of edges among v ’s neighbors and $c_v = 0$ when $d_v < 2$. Clustering and return probabilities are rounded to two decimals and printed in node order. These are explicit statistics, not learned graph embeddings.

3.3 Outcomes and paired analysis

We record **correct**, **wrong**, and **truncated** as distinct outcomes on every prompt. Truncations receive no correctness credit, and every reported correct or truncated share has all prompts in its denominator. Degree, node count, and edge count use integer exact match; cycle check uses Boolean exact match. For connected nodes, the primary score is Set-F1, $2|\hat{S} \cap S|/(|\hat{S}| + |S|)$, assigning zero to truncations and one to two correctly empty sets; exact set match is an additional diagnostic. For edge existence, we report raw accuracy with balanced accuracy $(\text{TPR} + \text{TNR})/2$ over all prompts and the yes-rate among *finished* responses. Truncations count as incorrect within their gold class when calculating balanced accuracy.

Every control–primer comparison pairs responses to the same graph and question. Exact McNemar tests use the two kinds of disagreement within those pairs: control correct/primer wrong, and the reverse (McNemar, 1947). Paired graph bootstraps provide intervals. Within each (model arm, task, control) family we adjust six primer

contrasts by Benjamini–Hochberg (Benjamini and Hochberg, 1995). The supplemental Set-F1 comparisons use paired random-sign tests with within-arm adjustment. We call an effect significant only at $q < .05$; effects at $q \geq .05$ are directions, and balanced-accuracy contrasts without corrected tests are descriptive. A graph-blind solver receives only the primer, checking whether it already supplies the answer. The main study records 84,000 responses on 700 distinct graph instances across task-specific density grids; Appendix A details coverage and provenance.

4 Results

Table 1 reports every main-sweep task, model arm, and primer: the no-primer correct share and the change on the *same* graph queries. All correct shares include truncated responses in the denominator. Asterisks identify BH-adjusted $q < .05$; unmarked changes are directions only. The table uses exact set match for neighbors and raw correctness for edge existence; their required primary and diagnostic measures appear in Table 5. Appendix A gives the complete truncation matrix (Table 6) and the two-task dense extension (Table 7); wrong share is the remainder after correct and truncated shares.

4.1 Answer availability does not ensure a gain

The graph-blind solver answers every queried degree from the degree primer and every edge count by summing degrees. Such gains demonstrate use of information already present in the preamble, not necessarily better inference from the incident encoding. Even for these directly disclosed answers, Table 1 shows opposite effects across arms: degree improves node-degree correctness in both 1.7B arms but reduces it in plain 4B; the latter also gains substantially on edge count. Near-ceiling unprimed performance limits the improvement available to thinking 4B. The three-feature bundle does not consistently outperform its constituents; in the dense plain-4B degree extension, clustering alone exceeds the bundle in a corrected paired contrast (Appendix B).

Effects also differ when the primer does not state the answer. The graph-blind solver recovers at most 0.8% of queried neighbor sets, yet the short component-count diagnostic improves plain-4B neighbor Set-F1 ($q < .001$); degree and the bundle reduce it ($q < .05$). For thinking 1.7B, de-

gree’s improvement in *exact* neighbor sets does not extend to primary Set-F1 ($q = .425$). Thus the metric used to assess a relational answer changes the conclusion. Table 5(a) gives every paired Set-F1 comparison, including the filler and component controls.

4.2 Density and prompt content modify the effect

At the same node count, the effect of a primer can change across densities. The dedicated degree follow-ups find that clustering helps plain 1.7B at lower densities, with a fresh-seed replication, but provides no supported gain in the denser group; a corrected exploratory interaction supports the difference between groups. For thinking 1.7B, the clustering direction reverses and degree’s benefit grows at higher densities, alongside more truncations (Appendix B). The groups contain different graphs, so density is a plausible moderator rather than an isolated cause. Early-listed queried nodes account for much of the plain clustering gain; node position and name length coincide in this setup. A structure-free filler also improves that task in the dedicated comparison, making an explanation based only on clustering’s topology untenable.

Constant-answer tasks offer a second warning. Every graph has 40 nodes and every sampled graph contains a cycle. Large changes on node count, including an improvement from filler, therefore reveal prompt sensitivity without testing whether the model distinguishes graph sizes. On cycle check, degree can lower correct share primarily by increasing truncation. Table 1 retains these tasks so negative and positive effects remain visible, but they cannot establish structural generalization.

4.3 Decisions, thinking, and completion

Edge existence changes require class-sensitive reading. In the main sweep, the bundle improves plain-1.7B raw and balanced accuracy as its finished-response yes-rate falls; RWSE has the opposite pattern for plain 4B. Table 5(b) gives raw accuracy, balanced accuracy, yes-rate, and truncation for each main-sweep condition. The dense plain-1.7B extension in Table 4 illustrates why raw accuracy alone is misleading: the unprimed arm almost always answers yes as positive-edge prevalence rises, whereas the bundle reduces this response bias and improves balanced accuracy descriptively. The main-sweep q -values test raw cor-

Task	Arm	None	Degree	Cluster.	RWSE	All	Comp.	Filler
Degree	1.7P	60.50	+7.5*	+3.0	+4.0	+10.2*	+1.8	-0.5
	1.7T	76.25	+8.8*	+1.5	+2.5	+12.2*	-1.0	-3.0
	4P	99.25	-6.5*	-0.5	-2.2*	-8.0*	+0.2	-0.5
	4T	97.00	+2.5	+1.8	-1.2	+0.5	+0.2	+2.0
Neighbors	1.7P	72.50	+3.5	+1.2	+1.0	+2.0	+0.8	-6.2*
	1.7T	78.75	+6.2*	+2.0	+0.2	+6.2*	+2.5	+2.5
	4P	86.00	-3.0	+1.0	+0.0	-2.8	+9.2*	-12.5*
	4T	95.50	+0.5	-0.8	+0.5	-1.2	+0.5	+1.2
Edge count	1.7P	1.25	+3.0 [†]	-0.5 [†]	-0.8 [†]	+0.8 [†]	-1.0 [†]	-0.5 [†]
	1.7T	15.75	-3.2 [†]	-1.3 [†]	-3.2 [†]	+5.0 [†]	-0.8 [†]	-1.5 [†]
	4P	2.00	+27.2*	-1.2	-1.5	+3.8*	-1.2	+1.2
	4T	20.00	+8.5* [†]	+24.8* [†]	+10.2* [†]	+1.8 [†]	+24.8* [†]	+12.2* [†]
Edge exists	1.7P	70.50	+4.0	+3.8	+5.0	+11.5*	-8.2*	-15.8*
	1.7T	99.00	-2.5	-1.0	-1.2	-1.8	-0.8	-0.5
	4P	90.50	-4.8*	+4.0*	-10.0*	-10.0*	-2.8	-7.0*
	4T	99.75	+0.0	-0.8	-1.2	-1.0	-0.5	+0.0
Node count	1.7P	31.50	+6.8*	+67.2*	+2.0	+47.0*	+21.2*	+39.8*
	1.7T	91.00	+5.5*	+9.0*	+9.0*	+9.0*	+1.5	+9.0*
	4P	90.25	+8.0*	+8.8*	+9.2*	+9.8*	+9.0*	+9.2*
	4T	99.50	+0.2	+0.2	+0.2	+0.5	+0.0	+0.5
Cycle	1.7P	90.75	-17.5* [†]	+8.5*	-0.2	+7.0*	+2.0	+3.0
	1.7T	97.00	-21.8* [†]	-0.5	+2.2*	-0.8	-7.5*	+3.0*
	4P	99.00	-2.2	+0.5	-0.2	-0.2	-2.0	-0.5
	4T	99.25	-5.5*	+0.0	+0.2	-0.5	-3.2*	+0.5

Table 1: Complete main-sweep correct shares: no-primer baseline (percent of all prompts) and paired primer changes (points) on the same 40-node graphs at $p \leq .50$. P/T: plain/thinking. * denotes $q < .05$ within the (arm, task, control) family; [†] flags at least 15% truncation on either side. Neighbor sets use exact-set correctness here; primary Set-F1 is in Table 5(a). Edge existence uses raw correctness here; its balanced accuracy and finished-response yes-rate are in Table 5(b). Node count and cycle have constant gold labels. Full truncation shares appear in Table 6. Values are rounded from the tagged [main] results.

rectness; balanced-accuracy contrasts are descriptive.

Thinking exposes a separate limit: finishing the query. Unprimed thinking 4B is correct on only 20.0% of edge-count prompts because 79.5% truncate, although 97.6% of *finished* answers are correct. A component-count preamble raises correct share mainly by lowering truncation, whereas the bundle reduces truncation while increasing wrong answers (Figure 1). Degree also raises thinking-1.7B truncations on degree extraction even where total correct share rises. These three-outcome shifts prohibit interpreting conditional accuracy over finished responses as the principal result. Finally, on matched degree and neighbor queries, the share on which *both* plain-4B answers are correct falls with degree ($q < .001$), despite its edge-count gain. The primer can improve one score while weakening cross-task consistency.

5 Discussion

The central pattern is selective usefulness, rather than a common benefit of making graph structure explicit. Graph models can learn from numerical structural encodings, but a language model must locate and use verbalized values amid the incident list and its own generated text. Directly answer-

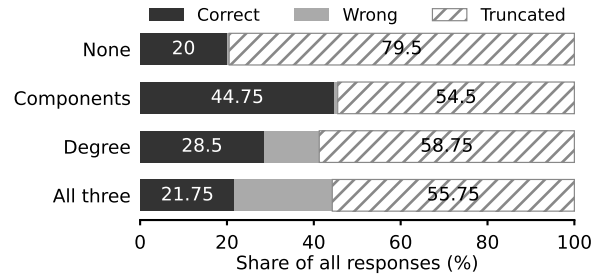


Figure 1: Qwen3-4B thinking on edge count: correct, wrong, and truncated shares of *all* responses. Conditions from top: none, component count, degree, all three node features. A lower truncated share need not imply a higher correct share.

bearing degree statements are useful evidence that the model sometimes accesses the preamble; they are *inference-time answer disclosure*, not training-set leakage or proof of graph traversal. Even that shortcut can fail when the baseline already counts accurately. The replicated clustering effect is smaller and cannot be explained by a primer-only solver, yet the filler and node-position patterns leave open a prompt-layout explanation. Output traces suggest changes in counting or retrieval strategies but cannot establish an internal mechanism.

Several axes interact. Density increases graph-

text complexity while changing edge prevalence; capacity sets baseline headroom; thinking changes the risk of exhausting the generation budget. Rounded RWSE has limited node-level resolution in denser graphs, while bundling multiple statistics increases text and can disrupt a useful single-feature effect. None of these observations justifies a universal ranking of primers. A fair comparison must report the task’s primary metric, the class-conditional decision pattern where relevant, and the correct/wrong/truncated shares over all assigned prompts. Paired graph comparisons reduce instance-level noise, but stochastic generation, preamble length and position, graph family, and unequal budgets across density extensions remain limitations. The fixed node count and uniformly positive cycle labels especially restrict what those tasks can demonstrate.

The findings also suggest a hierarchy of evidence for structural use. If the answer appears in the primer, increased correctness shows access to a useful textual fact. If only related facts appear, a paired gain is more interesting, but still admits explanations based on repeated node names, position, and altered generation length. Stronger evidence would require success on a query whose answer is absent from the preamble, together with checks that answers to related queries satisfy the same graph’s invariants. The joint degree-and-neighbor result points in this direction: a gain on a graph-level quantity can coexist with fewer pairs of mutually correct local answers. This distinction matters when transferring ideas from GraphGPS: providing a structural encoding does not guarantee that a frozen text model uses it as a graph model does.

The edge and truncation diagnostics illustrate a second general point about evaluation. At high edge prevalence, saying yes frequently can raise raw accuracy while leaving little ability to recognize a nonedge; balanced accuracy and yes-rate expose that pattern. On token-intensive counting tasks, finished generations can be accurate while most assigned prompts still fail. A primer can therefore change *what* is answered, *whether* an answer is finished, or both. Reporting each channel is necessary before interpreting an apparent gain as more reliable graph computation.

6 Future Work

Controlled follow-ups. First, test whether a primer still helps when it cannot directly disclose

the answer: omit queried-node values while preserving equivalent statistics for other nodes, and ask multi-step queries that require the incident edges. Second, randomize node labels and preamble order, and compare truly token-matched neutral text under equal generation budgets; these interventions would separate positional and length effects from graph information. Third, use variable-size and deliberately cyclic/acyclic corpora with balanced edge queries, then repeat the paired design on graph families beyond Erdős–Rényi. Finally, measure whether answers jointly respect $d_v = |N(v)|$ and $2|E| = \sum_v d_v$ across queries, not merely whether each answer matches in isolation. Record intermediate claims only as observable output, and test their consistency against the underlying graph; stated reasoning alone does not establish the procedure that produced a response.

Beyond these tests, our results suggest that a primer helps by doing computation the model cannot do reliably itself: stating the queried node’s degree raises thinking 1.7B by +8.8 points on node degree and lowers plain 4B, which already counts correctly, by −6.5. This points to six broader directions.

Who should compute. Compare four ways of giving a model the same structure: statistics computed in advance and placed in the prompt, graph tools the model can call, code it writes and runs, and the incident encoding alone. Varying model scale, graph size and task would show when structure should be computed for the model and when the model can work it out, a choice every system that pairs an LLM with graph data has to make.

Predicting primer benefit. Our effects track two measurable quantities: how much the primer reveals about the answer, which the graph-blind solver measures, and how poorly the model does without it. A model of the effect as the primer’s information gain times the model’s shortfall, minus a reading cost, could be fitted across many model families and tested on held-out models. The reading cost is substantial: in the dedicated degree follow-ups, filler costs thinking 1.7B 11.4 points at $p \geq .65$. Such a model also needs repeated sampling per query, because most primer effects sit on top of churn: in 89 of 120 (arm, task, primer) cells, a primer changes as many outcomes as filler does.

Graphs that do not fit. Beyond 40 nodes the encoding itself becomes the bottleneck. Under a

fixed token budget, which summary keeps the most queries answerable: statistics, sampled subgraphs, or the queried node’s neighborhood?

Searching the primer space. We chose four features and one wording by hand, and both matter. The plain-1.7B clustering gain is 5.01 points on queried nodes in the first third of the node list and -0.11 on the last third, and under RWSE, whose sentences state a return probability “after 2 steps”, 42.5% of thinking 4B’s correct cycle-check answers rest on a walk that is not a cycle (a two-node “cycle” or a retraced path), against 0.0% without a primer. An automated search over feature, wording, placement and scope, scored against length-matched and shuffled-value controls on held-out graphs, would bound what any primer can achieve.

Trust in stated statistics. Every primer we tested was correct. Among finished degree-primed node-degree responses that report a conflict between the stated degree and the model’s own count, thinking 4B ends on the stated value in 93%, thinking 1.7B in 66% and plain 4B in 56%. Primers corrupted at controlled rates would show whether models verify stated structure or copy it, which matters wherever LLMs rely on metadata that may be stale or wrong.

Mechanism. With open weights, activation patching could show whether a model uses a primer’s content or responds only to a longer, differently formatted prompt.

A controlled benchmark. The controls this study needed could be released as a benchmark, so that future claims that structure helps LLMs face the confounds we met: a graph-blind solver that exposes answer disclosure, truncation as its own outcome, length-matched content-free text, churn-aware paired comparisons, degree-preserving rewiring that changes clustering while fixing the degree sequence, gold answers and prompt length, and checks that correct answers rest on true claims. Without a primer, 43.5% of plain 1.7B’s correct cycle-check answers rest on an invented cycle.

7 Conclusion

Explicit graph statistics sometimes help a language model answer textual graph queries, but their effects depend on the task, model arm, density, and whether generation finishes. The clearest degree gains can be obtained directly from the primer;

answer-free and adjacency-related gains are narrower and sensitive to prompt content. Truncation and decision bias can reverse the impression left by a single accuracy figure. Consequently, verbalizing a feature that benefits trained graph architectures is a testable prompting strategy rather than a general recipe for better graph reasoning. Paired comparisons, primer-only checks, primary task metrics, and complete outcome reporting establish what improved; controlled counterfactual prompts and consistency tests are needed to establish why.

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A Evaluation Grid and Audit Trail

Coverage. Every result in the body uses an undirected graph with exactly 40 nodes. The main sweep uses four graph densities for each task; only degree and edge existence additionally use three denser levels. Each model arm sees the same graph instances and seven conditions, so the paired key is (task, instance, model arm, condition). Table 2 makes the task-specific coverage and scoring distinction explicit. A fixed-size node count and the observed all-positive cycle labels mean that those two tasks test prompt behavior rather than sensitivity to the graph.

Question	Densities	Reported score
Node degree	main + dense	Integer exact match
Connected nodes	main	Set-F1; exact set also
Edge count	main	Integer exact match
Node count	main	Integer exact match
Cycle check	main	Boolean exact match
Edge existence	main + dense	Raw, balanced, yes-rate

Table 2: Task coverage. “Main” denotes $p \in \{.10, .20, .35, .50\}$; “dense” denotes $p \in \{.65, .75, .85\}$. The yes-rate is calculated among finished responses.

Primer audit. Table 3 gives average preamble length in *characters*. Component count and filler are diagnostics outside the three-feature bundle. Filler names nodes without stating structure; it is approximately length-matched to clustering but differs in wording. Degree states the queried value and, by summation, edge count. A blind solver that reads only primer text identifies this answer disclosure.

Condition	Mean chars	Information
Components	37	Number of connected components
Degree	891	Degree for each node
Clustering	1,629	Local clustering for each node
RWSE	2,949	Two rounded return values per node
All three	4,691	Degree, clustering and RWSE
Filler	1,829	Node names, no graph statistic

Table 3: Mean preamble length in characters, not model tokens, over the main-sweep graphs. The no-primer control adds no preamble.

Edge-existence diagnostic. Table 4 compares the same prompts within each dense level. These

descriptive figures make the class-balance and decision-bias argument checkable: raw accuracy, balanced accuracy and yes-rate must be read together, since the proportion of actual edges changes with density.

p	Primer	Raw acc.	Bal. acc.	Yes-rate
.65	none	.680	.529	.980
.65	all	.760	.647	.900
.75	none	.770	.521	.990
.75	all	.860	.723	.880
.85	none	.860	.533	.990
.85	all	.890	.661	.940

Table 4: Plain Qwen3-1.7B on edge existence at high density. Raw and balanced accuracy use every prompt; yes-rate uses finished answers. These cells have no truncation. Values are descriptive, without a per-density BH claim.

Scoring and provenance. A generation that reaches its output cap is *truncated*, even if text before the cap resembles an answer. The three outcome shares have the same denominator: every graph prompt in that condition. Set-F1 likewise uses all prompts and scores a truncated response as zero; yes-rate alone is conditional on finishing. Exact McNemar tests compare paired correctness, with Benjamini–Hochberg q -values within each (arm, task, control) family. For the supplemental F1 analysis, we pair graph instances, use random-sign tests of the mean F1 difference and adjust the six primer contrasts within each arm; bootstrap intervals resample pairs. Only $q < .05$ is treated as significant. Cells with at least 15% truncation on either side are flagged and omitted from pooled low-truncation summaries, not from their own displayed results.

Tagged results come from GraphTalk’s results-sot, commit 292046f (GraphTalk Contributors, 2026): docs/results/n40-sweep.md and csv2/raw-trends/primer_findings.txt. scripts/build_raw_frame.py verifies saved gold labels; scripts/primer_findings.py rebuilds the tagged contrasts. Figure 1 uses [edgcount]; the bundled metric_audit.py and primary_metric_tables.py regenerate the primary-metric appendix from recorded responses.

(a) Neighbor-set Set-F1					(b) Edge-existence decisions					
Arm	Primer	None	With	Δ ; q	Arm	Primer	Raw	Bal.	Yes	Trunc.
1.7P	filler	0.9721	0.9624	-0.0096; 0.3114	1.7P	none	0.705	0.799	0.562	0.000
1.7P	components	0.9721	0.9716	-0.0004; 0.9503	1.7P	degree	0.745	0.823	0.517	0.000
1.7P	clustering	0.9721	0.9724	+0.0003; 0.9503	1.7P	clustering	0.743	0.824	0.525	0.000
1.7P	rwse	0.9721	0.9702	-0.0018; 0.9503	1.7P	rwse	0.755	0.827	0.501	0.003
1.7P	degree	0.9721	0.9794	+0.0074; 0.3352	1.7P	all	0.820	0.877	0.448	0.000
1.7P	all	0.9721	0.9626	-0.0094; 0.3846	1.7T	none	0.990	0.990	0.271	0.003
1.7T	filler	0.9732	0.9769	+0.0037; 0.8851	1.7T	degree	0.965	0.976	0.299	0.005
1.7T	components	0.9732	0.9726	-0.0006; 0.9942	1.7T	clustering	0.980	0.986	0.286	0.003
1.7T	clustering	0.9732	0.9830	+0.0098; 0.4254	1.7T	rwse	0.978	0.982	0.283	0.003
1.7T	rwse	0.9732	0.9577	-0.0156; 0.4254	1.7T	all	0.973	0.981	0.295	0.000
1.7T	degree	0.9732	0.9627	-0.0105; 0.4254	4P	none	0.905	0.932	0.357	0.000
1.7T	all	0.9732	0.9733	+0.0001; 0.9942	4P	degree	0.858	0.903	0.407	0.005
4P	filler	0.9871	0.9660	-0.0212; 0.0001	4P	clustering	0.945	0.963	0.323	0.000
4P	components	0.9871	0.9976	+0.0105; 0.0001	4P	rwse	0.805	0.867	0.463	0.000
4P	clustering	0.9871	0.9845	-0.0026; 0.5411	4P	all	0.805	0.867	0.461	0.003
4P	rwse	0.9871	0.9776	-0.0095; 0.0681	4T	none	0.998	0.998	0.268	0.003
4P	degree	0.9871	0.9768	-0.0103; 0.0379	4T	degree	0.998	0.998	0.268	0.003
4P	all	0.9871	0.9743	-0.0128; 0.0209	4T	clustering	0.990	0.993	0.270	0.010
4T	filler	0.9636	0.9720	+0.0084; 0.8279	4T	rwse	0.985	0.987	0.269	0.015
4T	components	0.9636	0.9641	+0.0006; 0.8956	4T	all	0.988	0.992	0.271	0.013
4T	clustering	0.9636	0.9544	-0.0092; 0.8279						
4T	rwse	0.9636	0.9676	+0.0041; 0.8279						
4T	degree	0.9636	0.9685	+0.0050; 0.8279						
4T	all	0.9636	0.9551	-0.0084; 0.8279						

Table 5: Primary-metric audit on the same 400 main-sweep prompts per arm and condition. (a) Neighbor-set Set-F1, with truncations scored zero: each row gives the no-primer baseline, the primer score and paired difference; q adjusts six contrasts per arm. Components and filler are diagnostics. (b) Edge-existence raw and balanced accuracy (Bal.) and truncation (Trunc.) use all prompts; yes-rate (Yes) uses finished answers. Compare each primer row with its arm’s none row. Only raw-accuracy contrasts have BH tests in Table 1; other edge measures are descriptive. P/T: plain/thinking.

Task	Arm	None	Degree	Cluster.	RWSE	All	Comp.	Filler
Degree	1.7P	0.00	+0.0	+0.0	+0.2	+0.0	+0.0	+0.0
	1.7T	0.75	+4.5	-0.5	+0.0	-0.5	-0.2	-0.5
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	2.75	-2.5	-2.0	-0.5	-1.0	-0.5	-1.8
Neighbors	1.7P	0.00	+0.0	+0.0	+0.0	+0.2	+0.0	+0.0
	1.7T	1.25	+0.2	-1.0	-0.8	-0.5	-0.3	-0.8
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	3.25	-0.3	+1.2	-0.3	+1.0	+0.3	-0.5
Edge count	1.7P	29.75	-20.2	-11.8	-11.8	-1.7	-2.5	-3.5
	1.7T	81.75	-0.7	+0.7	+5.0	-12.3	+0.5	+1.2
	4P	0.00	+1.8	+0.2	+0.5	+0.8	+0.8	+0.0
	4T	79.50	-20.8	-26.0	-12.3	-23.8	-25.0	-13.0
Edge exists	1.7P	0.00	+0.0	+0.0	+0.2	+0.0	+0.0	+0.0
	1.7T	0.25	+0.2	+0.0	+0.0	-0.2	+0.0	-0.2
	4P	0.00	+0.5	+0.0	+0.0	+0.2	+0.0	+0.0
	4T	0.25	+0.0	+0.8	+1.2	+1.0	+0.5	+0.0
Node count	1.7P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	1.7T	7.75	-5.0	-7.8	-7.8	-7.8	-0.5	-7.8
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	0.00	+0.2	+0.2	+0.2	+0.0	+0.0	+0.0
Cycle	1.7P	9.00	+8.2	-8.2	-7.2	-8.0	-2.5	-2.7
	1.7T	3.00	+21.2	+0.5	-2.2	+0.8	+7.5	-3.0
	4P	0.75	+0.8	-0.2	+0.5	-0.2	+0.8	+0.5
	4T	0.75	+4.8	+0.0	-0.2	+0.5	+3.2	-0.5

Table 6: Complete main-sweep truncated shares: no-primer baseline (percent of all prompts) and paired changes (percentage points), for the same queries as Table 1. Each of its cells’ *wrong* share equals 100% minus its correct and truncated shares. P/T: plain/thinking; components and filler are diagnostics. No significance marks are assigned to truncation contrasts.

(a) Correct share								
Task	Arm	None	Degree	Cluster.	RWSE	All	Comp.	Filler
Degree	1.7P	9.67	+0.3	-3.3	-5.3	+2.7	+0.3	-5.3
	1.7T	46.33	+25.7	-9.0	-4.3	+22.7	-1.7	-12.0
	4P	55.00	+9.3	+11.3	+7.7	-21.3	-1.0	+2.0
	4T	97.33	-0.7	-0.7	-0.7	+0.3	-2.0	+0.0
Edge exists	1.7P	77.00	+2.0	+1.3	+1.0	+6.7	-1.0	-1.3
	1.7T	99.00	-0.7	-0.3	+0.3	-0.7	+0.0	-0.7
	4P	90.00	-3.3	+5.3	-0.7	-2.3	-3.3	-6.3
	4T	99.33	+0.7	+0.7	+0.3	+0.3	+0.7	+0.7

(b) Truncated share								
Task	Arm	None	Degree	Cluster.	RWSE	All	Comp.	Filler
Degree	1.7P	0.33	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3
	1.7T	1.67	+9.0	-1.3	-1.3	+0.0	+0.3	-1.0
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	0.33	+0.3	+0.0	+0.7	+0.7	+0.3	-0.3
Edge exists	1.7P	0.00	+1.3	+0.0	+0.0	+0.0	+0.0	+0.0
	1.7T	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4P	0.00	+0.3	+0.0	+0.0	+0.3	+0.3	+0.0
	4T	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0

(c) Edge-existence decisions					
Arm	Primer	Raw	Bal.	Yes	
1.7P	none	0.770	0.527	0.987	
1.7P	degree	0.790	0.606	0.926	
1.7P	clustering	0.783	0.564	0.960	
1.7P	rwse	0.780	0.548	0.977	
1.7P	all	0.837	0.674	0.907	
1.7P	components	0.760	0.507	0.997	
1.7P	filler	0.757	0.500	1.000	
1.7T	none	0.990	0.993	0.747	
1.7T	degree	0.983	0.980	0.753	
1.7T	clustering	0.987	0.987	0.750	
1.7T	rwse	0.993	0.986	0.763	
1.7T	all	0.983	0.975	0.760	
1.7T	components	0.990	0.993	0.747	
1.7T	filler	0.983	0.984	0.747	

Arm	Primer	Raw	Bal.	Yes	
4P	none	0.900	0.832	0.803	
4P	degree	0.867	0.740	0.870	
4P	clustering	0.953	0.932	0.763	
4P	rwse	0.893	0.813	0.817	
4P	all	0.877	0.756	0.866	
4P	components	0.867	0.754	0.849	
4P	filler	0.837	0.697	0.873	
4T	none	0.993	0.996	0.750	
4T	degree	1.000	1.000	0.757	
4T	clustering	1.000	1.000	0.757	
4T	rwse	0.997	0.998	0.753	
4T	all	0.997	0.993	0.760	
4T	components	1.000	1.000	0.757	
4T	filler	1.000	1.000	0.757	

Table 7: Complete dense-extension comparison at $p \in \{.65, .75, .85\}$ for degree and edge existence. (a,b) No-primer correct/truncated share (percent of all 300 prompts), followed by paired primer changes (points); wrong share is the remainder. (c) Edge raw accuracy (Raw) and balanced accuracy (Bal.) include truncations as noncorrect; yes-rate (Yes) uses finished answers. All values are descriptive: no pooled or density-level significance is claimed, and plain-arm budgets differ from the lower-density sweep. P/T: plain/thinking.

B Screening and Focused Robustness Checks

Earlier screens. The repository’s `docs/results/README.md` indexes Gemma 4 E4B/12B and Qwen3-8B/14B, each in plain and thinking mode, on published 5–19-node GraphQA graphs; it reports near-ceiling unprimed performance on most piloted tasks. A separate no-primer 20/40/80-node screen has saved runs but *unverified* older analysis. It documents the selection effort without proving any size or model optimal; the final paired study uses only Qwen3-1.7B/4B and 40-node graphs.

Selection and replication. The clustering-on-degree condition emerged from a 40-cell screen: 700 dedicated graphs overlap it, while 2,100 do not. The independent fresh seed gives +4.2 points ($p = .00055$), with screen-wide Bonferroni-adjusted $p = .022$; the dedicated contrasts have separately reported BH q -values. Overlap limits the independence of the larger dedicated run.

Arm / primer	Low	High	High-low	q
1.7P / cluster.	+3.81	−0.67	−4.48	.00445
1.7T / cluster.	+2.50	−5.42	−7.92	< .001
1.7T / degree	+11.31	+24.75	+13.44	< .001

Table 8: Primer effects on degree correct share (points), each relative to no primer. Low: $p \leq .50$ (1,600 pairs); high: $p \geq .65$ (1,200 pairs). High-minus-low graph-bootstrap intervals are $[-7.62, -1.42]$, $[-11.75, -4.04]$, and $[+9.12, +17.71]$ points in row order. These three post hoc tests constitute the BH family.

Sensitivity. Split-half intermediate-headroom means remain +13.9 and +16.2 points for answer-carrying cells; they pool heterogeneous tasks and arms. The repository’s `[bandsens]` checks truncation thresholds. Early IDs are also short, confounding position with tokenization.

Reproduction. `robustness_audit.py` regenerates the tagged tests from saved responses; `robustness_audit.txt` contains their output. Screening and position use the repository’s tagged reports. No new model inference was performed.

Fixed-size effect heterogeneity. Table 8 compares within-graph primer effects between density groups, including truncations as noncorrect outcomes. Welch tests compare graph-level differences, density-stratified bootstrap supplies intervals, and BH adjusts the three exploratory interactions. Different graphs populate each density, so the test does not isolate its causal role.

Direct bundle contrast. Across 300 paired dense plain-4B degree queries, clustering alone is correct on 66.33% versus 33.67% with *all* (−32.67 points, 95% bootstrap interval $[-39.33, -25.67]$, $q < .001$ across six alternatives). Neither condition truncates. All six paired comparisons appear in `robustness_audit.txt`. Length and order remain confounded.